

Course: Cloud and Network Security

Student Name: Tamara Zawadi

Student Number: CS-WIC06-26028

Week 1: Network Standards, Models & Protocols

Assignment 2: Use Wireshark to Examine Network Traffic

Introduction

This lab involved using Wireshark, a network protocol analyzer, to capture and analyze network traffic. The focus was on Internet Control Message Protocol (ICMP) packets generated when using the ping command. The lab demonstrated how network packets travel between devices and how information such as source and destination IP addresses can be examined using Wireshark.

Objectives

Part 1: Capture and analyze local ICMP data in Wireshark

Part 2: Capture and analyze remote ICMP data in Wireshark

Requirements

1. Wireshark – To capture and analyze network packets
2. PC – To generate and capture network traffic
3. Additional PC (teammate's PC)– To reply to ping requests
4. Internet /LAN connection – To provide network connectivity

Part 1: Capture and analyze local ICMP data in Wireshark

Step 1: Retrieve your PC interface addresses

Type **ipconfig /all** into your command prompt to retrieve the IP address of your PC interface, its description and MAC address.

```
Command Prompt
Microsoft Windows [Version 10.0.26100.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>ipconfig/all

Windows IP Configuration

Host Name . . . . . : Zawadi
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (6) I219-LM
Physical Address. . . . . : F8-0D-AC-CD-0B-A6
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-20
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . . : fe80::f11a:59e2:4019:5ec8%32(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 84433878
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-B3-90-91-F8-0D-AC-CD-0B-A6
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
localdomain

Mobile Broadband adapter Cellular:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Generic Mobile Broadband Adapter
Physical Address. . . . . : 8E-9F-93-51-52-51
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

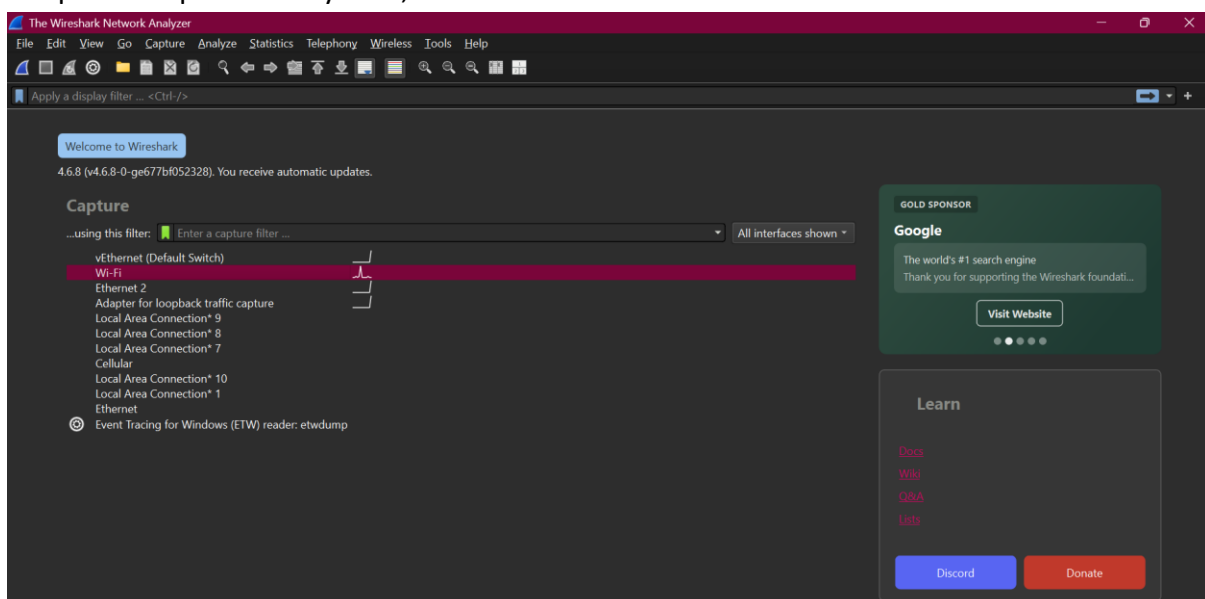
Ethernet adapter vEthernet (Default Switch):

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-38-CC-00
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . . : fe80::e396:4868:c195:5d0e%57(Preferred)
IPv4 Address. . . . . : 172.21.144.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 956306781
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-B3-90-91-F8-0D-AC-CD-0B-A6
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\user>
```

Step 2: Start Wireshark and begin capturing data

Navigate to Wireshark and double click the desired interface (make sure it has traffic) to start the packet capture. In my case, I went with Wi-Fi.



Information will start scrolling down the top section in Wireshark and data lines will appear in different colors based on the protocol.

We will apply the ICMP filter to make it easier to view and work with the data.

Type **icmp** in the filter box at the top of Wireshark and press enter to view only ICMP PDUs. This filter causes the data in the top window to disappear but traffic is still being captured on the interface.

Navigate to a command prompt window and ping the additional IP address received from a team member or an extra device. This device should be connected to the same network as your device.

```
Command Prompt
Physical Address. . . . . : 8E-9F-93-51-52-51
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter vEthernet (Default Switch):

Connection-specific DNS Suffix . : 
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-38-CC-00
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e396:4868:c195:5d0e%57(Preferred)
IPv4 Address. . . . . : 172.21.144.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 956306781
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-B3-90-91-F8-0D-AC-CD-0B-A6
NetBIOS over Tcpip. . . . . : Enabled

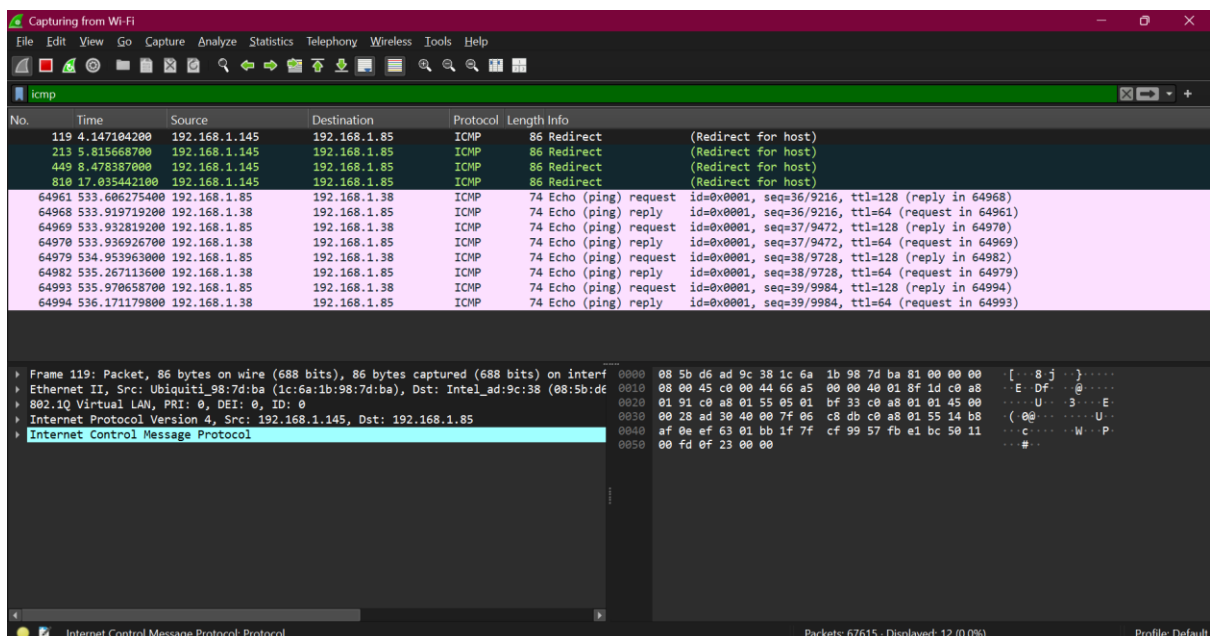
C:\Users\user>ping 192.168.1.38

Pinging 192.168.1.38 with 32 bytes of data:
Reply from 192.168.1.38: bytes=32 time=1647ms TTL=64
Reply from 192.168.1.38: bytes=32 time=4ms TTL=64
Reply from 192.168.1.38: bytes=32 time=313ms TTL=64
Reply from 192.168.1.38: bytes=32 time=200ms TTL=64

Ping statistics for 192.168.1.38:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 1647ms, Average = 541ms

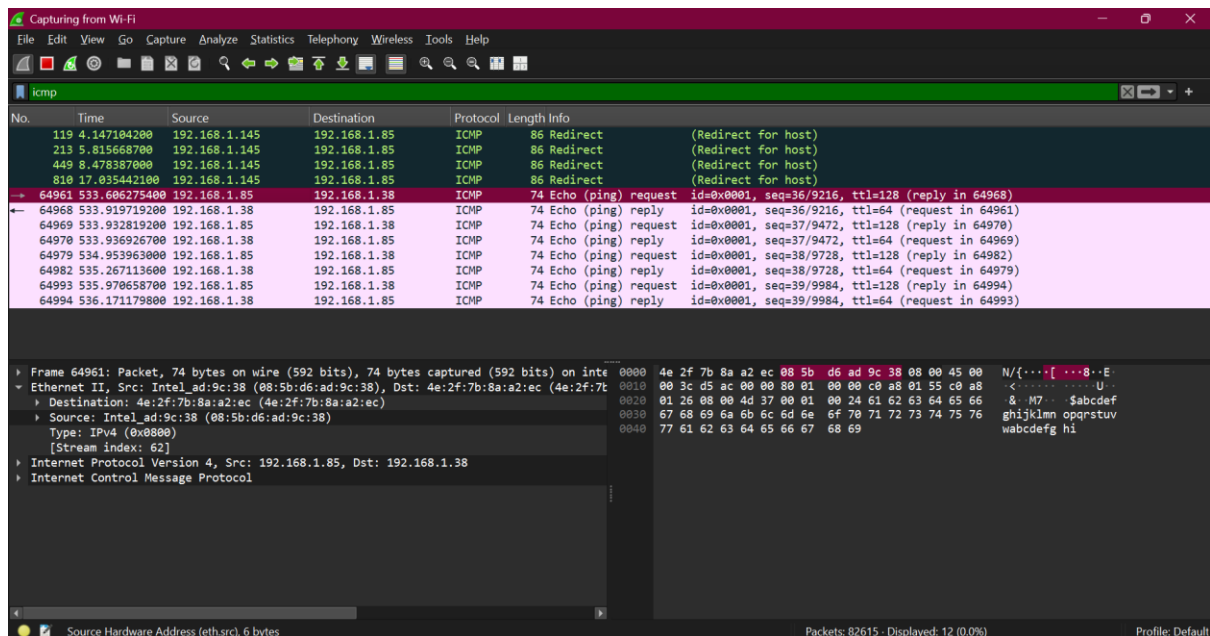
C:\Users\user>
```

Notice that data starts appearing on the top window of wireshark again.



Step 3: Examine captured data

Click the first ICMP request PDU frame from the ping. Notice that the source column has your IP address and the destination column has the IP address of the pinged device. This can be seen in detail on the Ethernet II section which shows the destination and source MAC addresses.



Note that the source MAC address matches that of your PC interface and the destination MAC address matches that of your teammate's PC

The MAC address of your teammate's PC is obtained by your PC using the ARP – Address Resolution Protocol. Your PC checks its ARP table to see if it already has the target IP address matched with a MAC address. If the MAC address is missing, your PC broadcasts an ARP request packet to every device on the local network asking, "Who has this IP address?". The target computer recognizes its own IP address and sends back an ARP reply containing its specific MAC address. Your PC saves this MAC address in its ARP cache for future use before sending out the actual ping data to the destination device.

Part 2: Capture and Analyze remote ICMP data in Wireshark

In this part, you will ping remote hosts and examine the generated data.

Step 1: Start capturing data on the interface

- Ping the following URLs from the Windows Command Prompt:
 1. www.yahoo.com

```
Command Prompt
Microsoft Windows [Version 10.0.26100.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>ping www.yahoo.com

Pinging me-ycpi-cf-www.g06.yahoodns.net [102.165.180.206] with 32 bytes of data:
Reply from 102.165.180.206: bytes=32 time=64ms TTL=55
Reply from 102.165.180.206: bytes=32 time=62ms TTL=55
Reply from 102.165.180.206: bytes=32 time=67ms TTL=55
Reply from 102.165.180.206: bytes=32 time=70ms TTL=55

Ping statistics for 102.165.180.206:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 62ms, Maximum = 70ms, Average = 65ms

C:\Users\user>
```

2. www.cisco.com

```
Command Prompt
Microsoft Windows [Version 10.0.26100.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>ping www.cisco.com

Pinging e2867.dsca.akamaiedge.net [2.17.168.94] with 32 bytes of data:
Reply from 2.17.168.94: bytes=32 time=11ms TTL=59
Reply from 2.17.168.94: bytes=32 time=27ms TTL=59
Reply from 2.17.168.94: bytes=32 time=13ms TTL=59
Reply from 2.17.168.94: bytes=32 time=15ms TTL=59

Ping statistics for 2.17.168.94:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 27ms, Average = 16ms

C:\Users\user>
```

3. www.google.com

```
Command Prompt
Microsoft Windows [Version 10.0.26100.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>ping www.google.com

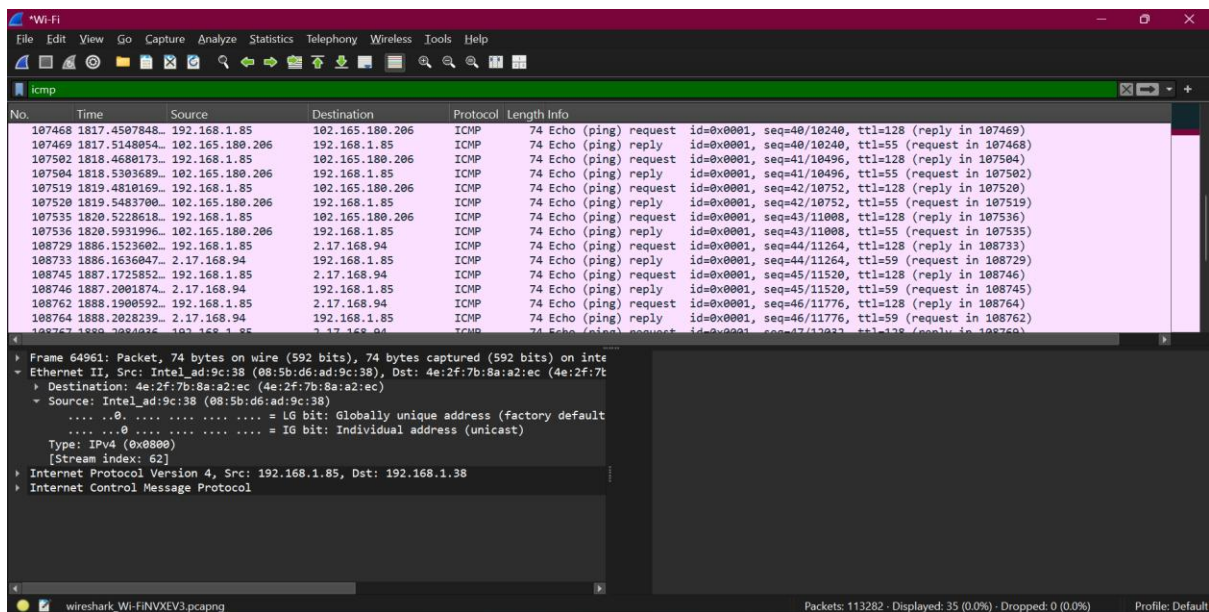
Pinging www.google.com [142.251.150.119] with 32 bytes of data:
Reply from 142.251.150.119: bytes=32 time=53ms TTL=118
Reply from 142.251.150.119: bytes=32 time=157ms TTL=118
Reply from 142.251.150.119: bytes=32 time=42ms TTL=118
Request timed out.

Ping statistics for 142.251.150.119:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 157ms, Average = 84ms

C:\Users\user>
```

Step 2: Examining and analyzing data from the remote hosts

Review the captured data in Wireshark and examine the IP and MAC addresses of the three locations that you pinged.



Observed IP and MAC addresses:

- IP address for www.yahoo.com : 102.165.180.206
- MAC address for www.yahoo.com: e0:63:da:c7:51:68
- IP address for www.cisco.com : 2.17.168.94
- MAC address for www.cisco.com : e0:63:da:c7:51:68
- IP address for www.google.com : 142.251.150.119
- MAC address for www.google.com: e0:63:da:c7:51:68

Conclusion

The captured destination MAC addresses for all three remote IP addresses are the same. This is because the MAC address shown is that of your local router or default gateway. Because every external website packet leaves your computer through the exact same local router interface, the destination MAC address on your local PC will be identical for all of them.

This information differs from that of the local ping information in that the destination MAC address in the local ping was that of the teammate's PC since both devices were in the same network. However, in the remote ping, the destination MAC address is that of the router since it is the device connecting your PC to the external network.

Reflection

Wireshark demonstrated that local and remote network communication use MAC addresses differently. When communicating with a device on the same local network, the destination MAC address corresponds to the local destination device. When communicating with a remote host, the computer sends the Ethernet frame to the MAC address of its local default gateway. The router then forwards the packet toward the remote destination. This demonstrates the relationship between Layer 2 addressing and Layer 3 IP routing.

